

JATOTH SHYAMKUMAR

Hyderabad, Telangana, India | ☎ +91 8555982032 | ✉ jatothshyamkumar@gmail.com

PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of experience in C and Embedded C, simulator development, GUI applications, software debugging, functional testing, data analysis, and technical documentation. Proficient in developing Mission Computer and Bus Monitor simulators using Qt Creator and Visual Studio with hands-on experience in MIL-STD-1553B communication, DDC device drivers, BC-RT configurations, real-time data monitoring, and Hardware-in-the-Loop (HIL) simulation.

TECHNICAL SKILLS

Languages: Programming in C, Embedded C.

GUI & Tools: Qt Creator, Visual Studio, MATLAB.

Protocols & Drivers: MIL-STD-1553B, RS-422 Serial Communication, DDC Device Drivers

Simulation & Logic: Mission Computer Simulator, Bus Monitor Simulator, HIL Simulation, Timers, Scheduling, DIP/DOPs

Testing & Docs: Functional Testing, Software Debugging, Code Traceability, Code Validation, ICD, SRS, STR, STP, VDD

PROFESSIONAL EXPERIENCE

Research Centre Imarat (RCI), DRDO

Hyderabad

Embedded Software Engineer - Contract

Feb 2022 – Present

- Developed and maintained GUI-based simulator applications using Qt Creator and Visual Studio for testing On-Board Computer (OBC) software.
- Implemented real-time parameter display, monitoring, and timer-based execution functions for time-critical operations.
- Configured DDC drivers for MIL-STD-1553B data transmission/reception and established Bus Controller–Remote Terminal (BC-RT) communications.
- Participated actively in RIG Test, Carriage Flight Trial, and Release Flight Trial execution phases.
- Conducted software code validation before RCMA and R&QA teams, ensuring strict code traceability and functional accuracy.
- Authored and managed lifecycle technical documentation including ICD, SRS, STR, STP, and VDD.

KEY PROJECTS

GUI for Mission Computer Simulator

Qt Creator, DDC Drivers, MIL-STD-1553B, Timers, HIL Simulation

Project: Mission Computer Simulator for testing OBC software.

- Collected and analyzed basic simulator requirements and developed the GUI using Qt Creator.
- Implemented real-time parameter display and monitoring.
- Integrated DDC drivers for data transmission and reception through the hardware interface.
- Implemented BC-RT communication configuration with the simulator acting as Bus Controller and OBC as Remote Terminal.
- Implemented timers for executing simulator functions at different time intervals.
- Implemented a six-point Launch Acceptance Region display for monitoring the real-time weapon-launch scenario.
- Supported Hardware-in-the-Loop Simulation.

GUI for Bus Monitors

Visual Studio, DDC Drivers, MIL-STD-1553B

Project: Bus Monitor simulator for monitoring MIL-STD-1553B data.

- Collected and analyzed basic requirements for Bus Monitor applications.
- Developed Telemetry Bus, EW Bus, and Avionics Bus Monitor simulators using Visual Studio.
- Integrated DDC drivers for receiving communication data.
- Implemented the simulator as a monitoring terminal with GUI-based data display.
- Supported analysis and monitoring of MIL-STD-1553B communication data.

EDUCATION

Swami Vivekananda Institute of Technology (JNTUH)

Hyderabad

M.Tech in Embedded Systems (71.81%)

2014 – 2018

Keshav Memorial Institute of Technology (JNTUH)

Hyderabad

B.Tech in Electronics & Communication Engineering (64.90%)

2011 – 2014

Sana Polytechnic College (SBTET)

Kodad

Diploma in Electronics & Communication Engineering (69.63%)

2008 – 2011